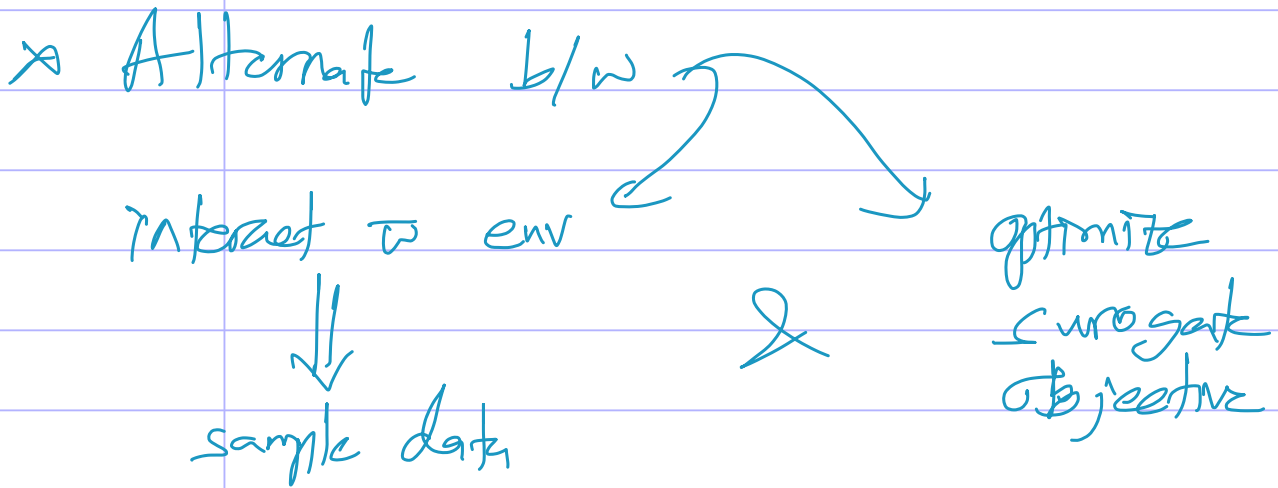


PPO



- * similar to TRPO but 1st order optimization

Policy Gradient Methods

- * compute an estimator of the policy gradient

- * optimize w/ SGD

- * Common gradient estimator:

$$\hat{g} = \hat{\mathbb{E}}_t \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \hat{A}_t \right]$$

where $\pi_\theta \rightarrow$ stochastic policy

$\hat{A}_t \rightarrow$ estimator of advantage
funcⁿ at time t

Class look at the gradient estimator

* Consider a supervised learning scenario
on a fixed dataset.

* Objective function

$$J(\theta) = \mathbb{E}_{u \sim \text{Dataset}} \left[\text{Loss}(f_\theta(u), y) \right]$$

This dataset is static

* dataset doesn't change when we
update network parameters θ

①

* So when we take gradient of objective func:-

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \left[\mathbb{E}_{x \sim \text{dataset}} [L_{\text{os}}(f_{\theta}(x), y)] \right]$$

$$= \mathbb{E}_{x \sim \text{dataset}} \left[\nabla_{\theta} L_{\text{os}}(f_{\theta}(x), y) \right]$$

by ①

* in RL setting the dataset itself moves with θ . So cannot do the above step of taking ∇ inside $\mathbb{E}$.

* consider RL objective
→ maximize expected reward over a trajectory T

$$J(\theta) = \mathbb{E}_{T \sim \pi_{\theta}} [R(T)] \quad \text{--- ②}$$

* here R function is computed by the environment. Not our NN

* Our NN parameters affect the probability of experiencing certain trajectories (states & reward)

* Rewarding ① a sum

$$J(\theta) = \sum_{\tau} P(\tau|\theta) R(\tau)$$

$$\nabla_{\theta} J(\theta) = \sum_{\tau} \nabla_{\theta} P(\tau|\theta) \cdot R(\tau) \quad \text{--- ③}$$

Cannot formulate as expectation.

* Usual expectation

$$\mathbb{E}_{\pi}[f(\pi)] \longleftrightarrow \sum_{\pi} P(\pi) \times f(\pi)$$

* but here we have

$$\sum_n \nabla p(n) \times f(n)$$

* Log derivative trick

$$\frac{d \log p(n)}{dn} = \frac{1}{n}$$

$$\frac{d \log p(n)}{dn} = \frac{1}{p(n)} \cdot \frac{dp(n)}{dn}$$

|||

$$\frac{d \log p(\tau|\theta)}{d\theta} = \frac{1}{p(\tau|\theta)} \cdot \frac{dp(\tau|\theta)}{d\theta}$$

i.e

$$\nabla_{\theta} \log p(\tau|\theta) = \frac{1}{p(\tau|\theta)} \cdot \nabla_{\theta} p(\tau|\theta)$$

rearranging

$$\nabla_{\theta} p(\tau|\theta) = p(\tau|\theta) \cdot \nabla_{\theta} \log p(\tau|\theta)$$

$$\nabla_{\theta} \log p(\tau|\theta)$$

— (4)

Use ④ in ①

$$\nabla_{\theta} J(\theta) = \sum_{\tau} \nabla_{\theta} P(\tau|\theta) \cdot R(\tau)$$

$$= \underbrace{\sum_{\tau} P(\tau|\theta) \cdot \nabla_{\theta} \log P(\tau|\theta)}_{\mathbb{E}_{\tau \sim \pi_{\theta}}} \cdot R(\tau)$$

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\nabla_{\theta} \log P(\tau|\theta) R(\tau) \right]$$

————— (*)

$$P(\tau|\theta) \longrightarrow \pi_{\theta}(a_t | s_t)$$

Environment Cancellation Trick

① Deconstruct the Trajectory

$$\tau \text{ is } (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$$

* $P(\tau|\theta)$ depends on 3 things

1. $P(s_0) \rightarrow$ prob. of starting at s_0

2. $\pi_\theta(a_t|s_t) \rightarrow$ prob. of agent choosing a_t in s_t

3. $P(s_{t+1}|s_t, a_t) \rightarrow$ prob. of state transition from $s_t \rightarrow s_{t+1}$ if we do a_t

i.e.

$$P(\tau|\theta) = P(s_0) \prod_{t=0}^T P(s_{t+1}|s_t, a_t) \pi_\theta(a_t|s_t)$$

② Apply Log

$$\log P(\tau|\theta) = \log P(s_0) + \sum_{t=0}^T \left[\log P(s_{t+1}|s_t, a_t) + \log \pi_\theta(a_t|s_t) \right]$$

— ①

② Take Gradient w.r.t θ

$$\nabla_{\theta_1}^{\log} P(s_0) = 0$$

initial state determined by env. not N.R

$\nabla_{\theta} \log P(S_{t+1} | S_t, a_t) = 0$ state transitions
are determined by
env. dynamics, not θ

$\nabla_{\theta} \log \pi_{\theta}(a_t | S_t) \neq 0$ this is
our policy network.

$\therefore \textcircled{1} \Rightarrow$

$$\nabla_{\theta} \log P(\tau | \theta) = \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | S_t) \quad \textcircled{2}$$

$\textcircled{1}, \textcircled{2} \Rightarrow$

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\left(\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | S_t) \right) \cdot R(\tau) \right]$$

$\Rightarrow$ Model free RL is possible!!!

i.e. we can update weights of our
model w/o knowing env dynamics!

$$\nabla_{\theta} J(\theta) = \sum_{t=0}^T \mathbb{E}_{\pi_{\theta}} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(t) \right] \quad \text{--- (3)}$$

* Causality of Reward. $\Rightarrow$

action at time t cannot affect reward at $t-1, t-2, \dots$

$$R(t) = \sum_{t'=0}^T r_{t'} = \underbrace{\sum_{t'=0}^{t-1} r_{t'}}_{\text{at time } = t \text{ this is constant.}} + \underbrace{\sum_{t'=t}^T r_{t'}}_{\text{this is variable.}} \quad \text{--- (4)}$$

Proof A

$$\begin{aligned} \mathbb{E}_{\pi_{\theta}} \left[\nabla_{\theta} \log p_{\theta}(a) \right] &= \sum_n p_{\theta}(a_n) \cdot \nabla_{\theta} \log p_{\theta}(a_n) \\ &= \sum_n \cancel{p_{\theta}(a_n)} \cdot \frac{1}{\cancel{p_{\theta}(a_n)}} \cdot \nabla p_{\theta}(a_n) \\ &= \sum_n \nabla_{\theta} p_{\theta}(a_n) \\ &= \nabla_{\theta} \sum_n \cancel{p_{\theta}(a_n)} \rightarrow 1 \end{aligned}$$

$\rightarrow 0$

$\rightarrow 5$

Use ④ in ⑤ $\leftarrow$ part 4

Consider

$$\mathbb{E}_{t \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t | s_t) R(t)]$$

$$= \mathbb{E}_{t \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a_t | s_t) \sum_{t'=0}^{t-1} r_{t'} \right]$$

$$+ \mathbb{E}_{t \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a_t | s_t) \sum_{t'=t}^T r_{t'} \right]$$

$$\left(\sum_{t'=0}^{t-1} r_{t'} \right) \cdot \mathbb{E}_{t \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a_t | s_t) \right] \rightarrow 0$$

$$\therefore \nabla J(\theta) = \sum_{t=0}^T \left[\mathbb{E}_{t \sim \pi_\theta} \left(\nabla_\theta \log \pi_\theta(a_t | s_t) \sum_{t'=t}^T r_{t'} \right) \right]$$

$\sum_{t'=t}^T r_{t'}$ has high variance.

So we use advantage estimator

$$A_t(a_t | s_t) = \sum_{t'=t}^T \gamma^{t'} r_{t'} - V_t(s_t)$$

$$\therefore \nabla_{\theta} J(\theta) = \sum_{t=0}^T \left[\mathbb{E}_{\substack{a_t \sim \pi_{\theta} \\ s_t \sim \pi_{\theta}}} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) A_t \right]$$

doesn't depend on
where T anymore. only on a_t & s_t

$$\therefore \nabla_{\theta} J(\theta) = \sum_{t=0}^T \left[\mathbb{E}_{s_t, a_t \sim \pi_{\theta}} \left(\log \pi_{\theta}(a_t | s_t) A_t \right) \right]$$

in practice, we let agent generate
 N samples & average over all the
steps in that sampling

$$\therefore \nabla_{\theta} J(\theta) \propto \hat{\mathbb{E}}_t \left[\log \pi_{\theta}(a_t | s_t) A_t \right]$$

shifting from theoretical expectations to
practical batch of data.